

THE ELEMENT-ORDER COLLISION PROBABILITY DISTINGUISHES THE 47 GROUPS OF ORDER 120

[AUTHOR NAME]

ABSTRACT. For a finite group G , let $N_k(G)$ denote the number of elements of order k in G , and define the *element-order collision probability*

$$P_2(G) = \sum_{k \geq 1} \left(\frac{N_k(G)}{|G|} \right)^2.$$

This is the Rényi-2 collision entropy of the order-type distribution viewed as a probability mass function on $\mathbb{Z}_{>0}$. We show by direct enumeration over the GAP `SmallGroups` library that the 47 pairwise non-isomorphic groups of order 120 have 47 pairwise distinct values of P_2 , so P_2 is a complete invariant on this order. Among these 47 groups, 27 ordered pairs (G_1, G_2) satisfy the *single-bucket-swap* property: their order-type distributions differ in exactly two orders k_1, k_2 by the same count $n \geq 1$, equivalently, passing from G_1 to G_2 redistributes n elements from order k_1 to order k_2 . The pair $(S_5, \text{SL}(2, 5))$ realises this pattern with $n = 24$, $k_1 = 2$, $k_2 = 10$, giving the explicit gap $P_2(S_5) - P_2(\text{SL}(2, 5)) = 1/300$. All claims are reproducible from a self-contained GAP script of fewer than 50 lines.

1. INTRODUCTION

The multiset of element orders of a finite group G , often called the *order type* of G [6], has been intensively studied as a coarse but information-rich invariant. Thompson asked whether sameness of order type forces sameness of solvability; the question was answered negatively by Piwek [5] in 2024. Cameron and Dey [2] place the order type in a partial order and study its combinatorial structure. The sum of element orders $\psi(G) = \sum_{x \in G} |x|$ is a scalar derived from the order-type distribution and has accumulated a substantial literature since Amiri, Jafarian Amiri and Isaacs [1]; see also [4] and references therein.

Less attention has been paid to the second moment of the order-type distribution, equivalently the *collision probability*

$$P_2(G) = \Pr_{(x,y) \in G^2} [|x| = |y|] = \sum_{k \geq 1} \left(\frac{N_k(G)}{|G|} \right)^2,$$

Date: May 17, 2026.

2020 Mathematics Subject Classification. Primary 20D60; Secondary 20E45, 05A18.

Key words and phrases. Finite groups, element orders, order type, group invariants, groups of order 120.

where the probability is over two uniformly independent draws from G and $|x|$ denotes the order of x . This is the standard collision probability associated with the order-type distribution, equivalently the Rényi entropy of order 2,

$$H_2(\text{order}(G)) = -\log_2 P_2(G).$$

The aim of this note is to record a small but striking computational fact: on the order $n = 120$, the scalar P_2 is sharp enough to distinguish all 47 pairwise non-isomorphic groups, and the pairs of groups whose P_2 values differ by a particularly small rational gap admit a clean structural description.

2. DEFINITIONS AND NOTATION

Let G be a finite group with $|G| = n$. For $k \in \mathbb{Z}_{>0}$ write

$$N_k(G) = \#\{x \in G : |x| = k\}.$$

Then $\sum_k N_k(G) = n$ and the *order-type* of G is the multiset $\{|x| : x \in G\}$, equivalently the function $k \mapsto N_k(G)$.

Definition 1. The *element-order collision probability* of G is

$$P_2(G) = \sum_{k \geq 1} \left(\frac{N_k(G)}{|G|} \right)^2 = \frac{1}{|G|^2} \sum_k N_k(G)^2.$$

Equivalently $P_2(G)$ is the probability that two uniformly random elements of G have the same order. The denominator $|G|^2$ implies that all P_2 values for groups of order n are rationals with denominator dividing n^2 .

Definition 2. Two finite groups G_1, G_2 with $|G_1| = |G_2|$ form a *single-bucket-swap pair* with parameters (k_1, k_2, n) if the difference of their order-type distributions is supported on exactly $\{k_1, k_2\}$ with

$$N_{k_1}(G_2) - N_{k_1}(G_1) = -n \quad \text{and} \quad N_{k_2}(G_2) - N_{k_2}(G_1) = +n,$$

i.e. passing from G_1 to G_2 moves n elements from order k_1 to order k_2 and leaves every other order count unchanged.

In a single-bucket-swap, the gap in P_2 has the closed form

$$(1) \quad P_2(G_2) - P_2(G_1) = \frac{1}{|G|^2} \left[(N_{k_2}(G_1) + n)^2 - N_{k_2}(G_1)^2 + (N_{k_1}(G_1) - n)^2 - N_{k_1}(G_1)^2 \right].$$

3. MAIN COMPUTATIONAL RESULTS

3.1. Distinct values on order 120.

Theorem 3. *There are 47 non-isomorphic groups of order 120. Their 47 values of $P_2(G)$ are pairwise distinct.*

Proof. A direct enumeration over `AllSmallGroups(120)` in GAP [3] computes each $N_k(G)$ from the conjugacy class table, then $P_2(G)$ as an exact rational. The 47 values, sorted ascending, are listed in Table 1. They are pairwise distinct. The reproducibility script is given in Section 5. $\square$

Remark 4. Theorem 3 should not be over-interpreted: P_2 is *not* a complete invariant for finite groups in general. Already at order 16 the order-type alone fails to distinguish, since $\mathbb{Z}_4 \times \mathbb{Z}_4$ and $\mathbb{Z}_2 \times Q_8$ share the same multiset $\{1, 3, 12\}$ (one identity, three involutions, twelve elements of order 4), hence the same $P_2 = (1 + 9 + 144)/256 = 154/256 = 77/128$. So the smallest order where P_2 fails to separate non-isomorphic groups is ≤ 16 , and Theorem 3 is a coincidence of order 120, not a structural fact.

3.2. Single-bucket-swap pairs at order 120.

Proposition 5. Among the $\binom{47}{2} = 1081$ unordered pairs of groups of order 120, exactly 27 are single-bucket-swap pairs.

Proof. By direct enumeration; see Table 2 and the GAP script. $\square$

Example 6 (S_5 vs $\text{SL}(2, 5)$). The two well-known non-abelian non-solvable groups of order 120 that contain A_5 as a central or quotient factor are S_5 and $\text{SL}(2, 5)$ (the binary icosahedral group, double cover of A_5). Their order-type distributions, computed from their conjugacy class tables, are:

$$S_5 : N_1 = 1, N_2 = 25, N_3 = 20, N_4 = 30, N_5 = 24, N_6 = 20,$$

$$\text{SL}(2, 5) : N_1 = 1, N_2 = 1, N_3 = 20, N_4 = 30, N_5 = 24, N_6 = 20, N_{10} = 24.$$

The two distributions differ in exactly two orders: order 2 (where S_5 has 25 elements and $\text{SL}(2, 5)$ has only the unique central involution) and order 10 (which has no representatives in S_5 but 24 in $\text{SL}(2, 5)$). Passing from $\text{SL}(2, 5)$ to S_5 redistributes 24 elements from order 10 to order 2. By (1),

$$P_2(S_5) - P_2(\text{SL}(2, 5)) = \frac{1}{120^2} [25^2 - 1^2 + 0^2 - 24^2] = \frac{624 - 576}{14400} = \frac{48}{14400} = \frac{1}{300}.$$

3.3. The full P_2 value table.

Table 1: All 47 groups of order 120 sorted by $P_2(G)$ ascending.
“ k ” is the index in the GAP `SmallGroups` library at order 120.

k	Structure	P_2
17	$C_{12} \times D_{10}$	181 / 1440
4	C_{120}	187 / 1440
20	$C_3 \times ((C_{10} \times C_2) \rtimes C_2)$	43 / 288
31	$C_{60} \times C_2$	221 / 1440
22	$C_{20} \times S_3$	391 / 2400
10	$C_3 \rtimes (C_4 \times D_{10})$	1181 / 7200
27	$C_4 \times D_{30}$	83 / 480
2	$C_3 \times (C_5 \rtimes C_8)$	251 / 1440
32	$C_{15} \times D_8$	17 / 96
11	$C_3 \rtimes ((C_{10} \times C_2) \rtimes C_2)$	1303 / 7200
8	$(C_3 \rtimes C_4) \times D_{10}$	1321 / 7200

(continued on next page)

k	Structure	P_2
6	$C_3 \times (C_5 \rtimes C_8)$	89 / 480
25	$C_5 \times ((C_6 \times C_2) \rtimes C_2)$	1343 / 7200
18	$C_3 \times D_{40}$	271 / 1440
40	$C_6 \times (C_5 \rtimes C_4)$	277 / 1440
13	$C_3 \rtimes ((C_{10} \times C_2) \rtimes C_2)$	1403 / 7200
15	$C_5 \times SL(2, 3)$	1411 / 7200
30	$C_3 \rtimes ((C_{10} \times C_2) \rtimes C_2)$	283 / 1440
19	$C_6 \times (C_5 \rtimes C_4)$	19 / 96
5	$SL(2, 5)$	1427 / 7200
36	$S_3 \times (C_5 \rtimes C_4)$	1429 / 7200
34	S_5	1451 / 7200
16	$C_3 \times (C_5 \rtimes Q_8)$	97 / 480
35	$C_2 \times A_5$	1457 / 7200
12	$C_3 \rtimes D_{40}$	1471 / 7200
1	$C_5 \times (C_3 \rtimes C_8)$	493 / 2400
9	$S_3 \times (C_5 \rtimes C_4)$	1493 / 7200
43	$C_{10} \times A_4$	1513 / 7200
37	$C_5 \times S_4$	1547 / 7200
33	$C_{15} \times Q_8$	323 / 1440
38	$C_5 \rtimes S_4$	1619 / 7200
14	$C_3 \times (C_5 \rtimes Q_8)$	1631 / 7200
24	$C_{10} \times (C_3 \rtimes C_4)$	1649 / 7200
23	$C_5 \times D_{24}$	187 / 800
39	$A_4 \times D_{10}$	563 / 2400
44	$C_2 \times C_6 \times D_{10}$	23 / 96
42	$C_2 \times S_3 \times D_{10}$	1813 / 7200
21	$C_5 \times (C_3 \rtimes Q_8)$	629 / 2400
3	$C_3 \rtimes (C_5 \rtimes C_8)$	137 / 480
28	D_{120}	47 / 160
47	$C_{30} \times C_2 \times C_2$	85 / 288
7	$C_3 \rtimes (C_5 \rtimes C_8)$	427 / 1440
26	$C_3 \rtimes (C_5 \rtimes Q_8)$	29 / 96
41	$C_2 \times (C_{15} \rtimes C_4)$	437 / 1440
29	$C_2 \times (C_{15} \rtimes C_4)$	89 / 288
45	$C_2 \times C_{10} \times S_3$	2261 / 7200
46	$C_2 \times C_2 \times D_{30}$	481 / 1440

Table 2: All 27 single-bucket-swap pairs at order 120, sorted first by n (number of elements moved) and then by denominator of the P_2 gap. Each row records that passing from group A to group B moves n elements from order k_1 to order k_2 .

A	B	n	k_1	k_2	$ P_2(A) - P_2(B) $
8	41	24	20	4	$3/25$
5	34	24	10	2	$1/300$
10	14	30	2	4	$1/16$
8	12	30	4	2	$1/48$
9	13	30	4	2	$1/80$
29	30	30	4	2	$9/80$
30	46	30	4	2	$11/80$
5	35	30	4	2	$1/240$
11	42	30	4	2	$17/240$
26	27	30	4	2	$31/240$
27	28	30	4	2	$29/240$
2	3	40	24	8	$1/9$
6	7	40	24	8	$1/9$
19	29	40	12	4	$1/9$
40	41	40	12	4	$1/9$
16	26	40	12	4	$1/10$
18	28	40	6	2	$19/180$
44	46	40	6	2	$17/180$
1	3	48	40	8	$2/25$
21	26	48	20	4	$1/25$
24	29	48	20	4	$2/25$
23	28	48	10	2	$3/50$
45	46	48	10	2	$1/50$
29	46	60	4	2	$1/40$
3	26	60	8	4	$1/60$
3	28	60	8	2	$1/120$
26	28	60	4	2	$1/120$

4. DISCUSSION

The result of Theorem 3 is a discrete coincidence: at order 120, the 47-element image of P_2 in $\mathbb{Q}$ happens to be injective. There is no reason to expect this on every order. Indeed, the order-type itself fails to be a complete invariant from order 16 onward, and any function of the order-type inherits this limitation in the worst case.

A natural question is to characterise the orders n at which P_2 is injective on the set of isomorphism classes. Computer experiments accessible to a

workstation can answer this for $n \leq 256$ or so; we leave the systematic computation, and the question of structural reasons (if any) for injectivity at specific orders, to follow-up work.

The single-bucket-swap pattern of Proposition 5 is a rigidity condition: most pairs of groups of order 120 differ in three or more order-buckets. The 27 that differ in exactly two are a small minority and tend to occur between groups that are structurally close (sharing a normal subgroup, or being central extensions of the same quotient by isomorphic kernels). Whether “single-bucket-swap” admits an algebraic characterisation in terms of standard group-theoretic relations is open.

5. REPRODUCIBILITY

All numerical claims are reproducible from the GAP script `reproducibility.g`, which we list in full below. On a workstation the entire computation runs in under one second.

```
LoadPackage("smallgrp");

n := 120;
num := NumberSmallGroups(n);
results := [];

for i in [1..num] do
  G := SmallGroup(n, i);
  desc := StructureDescription(G);
  ccs := ConjugacyClasses(G);
  orders := List(ccs, c -> Order(Representative(c)));
  sizes := List(ccs, Size);
  distinct_orders := Set(orders);
  Nk := [];
  for k in distinct_orders do
    nk := Sum(Filtered([1..Length(orders)], j -> orders[j] = k),
              j -> sizes[j]);
    Add(Nk, [k, nk]);
  od;
  p2 := Sum(Nk, x -> (x[2]/n)^2);
  Add(results, rec(id := i, desc := desc, Nk := Nk, p2 := p2));
od;

# Theorem: distinct P_2
distinct_p2 := Set(List(results, r -> r.p2));
Print("Distinct P_2 count: ", Length(distinct_p2), "\n");

# Proposition: 27 single-bucket-swap pairs
swap := 0;
```

```

for i in [1..num] do for j in [(i+1)..num] do
  a := results[i].Nk;; b := results[j].Nk;;
  all_k := Union(List(a, x -> [x[1]]), List(b, x -> [x[1]]));
  diff := [];
  for k in all_k do
    na := First(a, x -> x[1] = k); if na = fail then na := [k, 0]; fi;
    nb := First(b, x -> x[1] = k); if nb = fail then nb := [k, 0]; fi;
    if na[2] <> nb[2] then Add(diff, [k, nb[2] - na[2]]); fi;
  od;
  if Length(diff) = 2 and diff[1][2] + diff[2][2] = 0 then
    swap := swap + 1;
  fi;
od; od;
Print("Single-bucket-swap pair count: ", swap, "\n");

```

A worked Python post-processing script (parsing the GAP output, sorting the P_2 values, and listing the 27 swap pairs) is included in the supplementary materials accompanying this preprint.

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[AFFILIATION]

Email address: [email@example.org]